MA491 Project 2 Group Report

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1 Introduction

One of the characters in the television series *Game of Thrones* raises three dragons, and we determine whether it is feasible for these three dragons to survive on Earth by modeling their behavior, habits, diet, and interaction with the environment. We research the *Game of Thrones* dragons to model their energy expenditure and caloric intake. We also consider where the dragons could live and any impacts they would have on the environment. In *Game of Thrones*, humans support the dragons, so we determine the number of humans needed to support the dragons, with varying levels of support. Even though this model does not directly apply to a real situation, the model uses ideas for modeling real situations and could provide insight into the effects of introducing a predator into a new environment.

2 Model Overview

We model the energy expended by dragons from regular activities, such as flying and growing, as well as the energy they gain from eating. We also model the prey and dragon populations over time, including the effects of hibernation, reproduction, and migration. Since dragons continue to grow if they have enough food, the dragon population is limited by the amount of prey available at a location.

We use the following approach to find out what would happen if the three dragons from *Game of Thrones* lived in the Andes mountains and only ate llamas:

1. Iterate through the following steps for each day to predict what happens to the dragon and llama populations over time.
   (a) Compute the energy expended for metabolism using the current weight.
   (b) Compute the caloric intake and number of llamas to eat.
   (c) Based on the llama population, decide to hibernate, give birth, or migrate.
   (d) Compute the change in weight, dragon population, and llama population.
2. Analyze the results to determine the populations of dragons and llamas.

3 Assumptions

We assume:

1. Dragons only eat juvenile llamas. Juvenile llamas are llamas that are younger than two years old [4]
2. Llamas in South America only live in the Andes.
3. Llamas are evenly distributed in the Andes.
4. Dragons are not aggressive towards humans.
5. Dragons live in the same cave their entire adult lives.

6. Dragons use benzene to breathe fire.
7. The fire from the benzene burns off all of the carcinogenic material such that it does not cause cancer in the supporting humans.
8. Dragons only breathe fire to cook llamas.
9. Dragon eggs hatch immediately.
10. Dragons weigh 10 kilograms at birth.
11. Dragons weigh 35 kilograms after one year.
12. Dragons continue to grow with the proper dietary conditions.
13. The llama population is healthy.
14. One human can raise llamas living on 13,000 acres assuming that humans can travel roughly 10 times the land area of Rose-Hulman to take care of the llamas.
15. The llamas always have enough food.
16. Dragons are warm-blooded [17].
17. To reproduce, dragons spend one day forming the egg before laying it.
18. The energy required to produce an egg does not change with age.
19. The carrying capacity of llamas is 0.0272 llamas per acre. [4]
20. Dragons die if their weight falls below 20% of their weight before hibernation.

4 Location

Dragons prefer to live near volcanoes, so we find locations near volcanoes with sufficient prey to support their energy requirements [6]. From volcanic locations in [12], we decide that the dragons should live in the Andes, where there are seven million llamas total [13]. Dragons can only eat llamas in this model, but an expanded model could consider dragons flying to other volcanic locations with alternate prey sources, such as the mountains in Central America.

5 Dragon Behavior

Modeling how human interactions and violence affect dragon population is complex, and we assume dragons are not aggressive towards humans, though several other behaviors must be considered. Dragons fly each day to capture, burn, and eat llamas. Let v be a dragon's flight speed in kilometers per hour, and let $t_{h,\text{flying}}$ be the number of hours a dragon spends flying each day. Then, the maximum distance in kilometers a dragon can travel in one day is $vt_{\text{flying},h}$. If dragons have flight energy remaining after expending energy to get food, then they fly near their living location to maintain physical strength.

Dragons eat to maintain their weight as outlined in subsequent sections. If there are a sufficient number of llamas within $vt_{\text{flying},h}$ kilometers of flight, then the dragon eats those llamas. Since $vt_{\text{flying},h}$ kilometers represents the maximum distance flown away from and back to the dragon's cave, they can fly a maximum of $\frac{vt_{\text{flying},h}}{2}$ kilometers from their cave. If the number of llamas within a $\frac{vt_{\text{flying},h}}{2}$-kilometer radius of the dragon is below the number required to maintain weight, then the dragon recognizes that the environment cannot meet its energy needs and hibernates in a volcanically-warmed location. Dragon's cease hibernating after $t_{\text{hibernate}}$ years of rest. If the llama population is still too low and the dragon's weight drops below 20% of its weight before hibernation, the dragon dies and leaves the model.

A dragon can reproduce once it reaches a certain weight W_r, with reproduction depending on the amount of food in its location. If there are enough llamas to support two dragons, then another dragon will hatch from an egg in flames, as specified by *Game of Thrones* lore. We assume the egg hatches immediately for simplicity. The parent dragon does not carry any more children until at least one year after the birth.

During the first year of a baby dragon's life, the parent dragon hunts for the baby. Then, after the first year, the baby begins to hunt for itself while remaining with its parent to learn survival skills. A dragon can live on its own after one year, and the parent dragon can have another child if conditions support reproduction. When a dragon is seven years old, it becomes an adult and moves to a new location with more food, unless there are no other adults to care for hatchlings in its territory.

6 Energy Expenditure

Dragons are either hibernating or not hibernating, and hence, there are two cases for energy expenditure. Dragons also spend extra energy during reproduction.

6.1 Not Hibernating

Dragons use energy to gain weight, fly, breathe fire, and perform other standard metabolic activities. We suppose that younger dragons use a higher percentage of energy for gaining weight, with that percentage being near 50%. The fraction of total energy expended to gain weight is

$$P_{\text{weight}}(t) = -\frac{\frac{46.4}{1+e^{-1/200(t-365)}} + 56.4}{100}.$$

The numbers in this equation consider two horizontal asymptotes, the 50% maximum and the 10% minimum percentages of energy for gaining weight, as well as the energy requirements of young babies when cared for by their mothers. The equation creates a plot that approaches the 50% maximum at zero years of age for a baby. The plot slopes with a concave down shape until reaching an inflection point at 1 year old (10 kg). This is the point at which the parent dragon no longer needs to hunt and significantly care for the baby. After this point, the plot continues concave up, approaching the 10% asymptote as time continues.

The energy a dragon expends each hour for standard metabolic processes depends on its weight, $W(t)$, with t being time since birth in days [1],

$$E_{\text{standard metabolism}}(t) = 10^{\log(3.1)+0.744\log(W(t))},$$

and the energy it expends each hour while flying is

$$E_{\text{flying}}(t) = 10^{\log(37.152)+0.744\log(W(t))}.$$

Since $t_{\text{flying},h}$ describes the number of hours each day spent flying, the energy required for a dragon's daily activities is $(24 - t_{\text{flying},h}) \cdot E_{\text{standard metabolism}} + t_{\text{flying},h} \cdot E_{\text{flying}}$.

Dragon fire lights under a spontaneous reaction, meaning that the energy released from burning benzene causes more of the released benzene to burn. Therefore, the dragons only need to start fires to cook llamas. The energy required to start a fire is 0.1 mJ [3], which is negligible compared with other energy expenditures. Ignoring the ignition energy of the fire gives the energy a dragon expends in one day as

$$E_{\text{expended}}(t) = \frac{1}{1 - P_{\text{weight}}(t)} \left((24 - t_{\text{flying},h}) \cdot 10^{\log(3.1)+0.744\log(W(t))} + t_{\text{flying},h} \cdot 10^{\log(37.152)+0.744\log(W(t))} \right).$$

6.2 Hibernating

Energy expenditure decreases during hibernation. No flying or fire breathing occurs during hibernation, and the dragon's heating requirements reduce, especially considering the hibernation locations: caves warmed by

volcanoes. Due to these energy decreases, dragons exhibit about 1.2% of the standard metabolic rate during hibernation for total energy expenditure,

$$E_{\text{hibernate}}(t) = 0.012 \cdot E_{\text{standard metabolism}}(t).$$

6.3 Energy Expenditure through Reproduction

Dragons expend energy when producing eggs. According to [2], the energy used to form and lay an egg costs about 22% of an animal's standard metabolic rate, so

$$E_{\text{reproduce}} = 0.22 \cdot E_{\text{standard metabolism}}.$$

The energy required to lay an egg does not change with age or mass. Dragons with a mass of W_r kg can reproduce, meaning the following equation portrays the total energy required for a dragon at any valid reproductive age to lay an egg:

$$E_{\text{reproduce}} = 0.22 \cdot 10^{\log(3.1) + 0.744 \log(W_r)}.$$

Since dragons only carry eggs for a single day, the total expenditure is $E_{\text{reproduce}}$ kcal.

7 Weight

Similarly to energy expenditure, there are two cases for weight because a dragon gains weight while not hibernating and loses weight while hibernating.

7.1 Not Hibernating

Since change in weight is proportional to the amount of energy used to gain weight, we have a general model

$$\frac{dW}{dt} = \alpha E_{\text{weight}},$$

with $W(t)$ describing the dragon's weight in kg, t being time in days, α being a constant, and E_{weight} being energy expended to gain weight.

We know that $E_{\text{weight}} = P_{\text{weight}} \cdot E_{\text{expended}}$, so

$$E_{\text{weight}}(t) = \frac{P_{\text{weight}}(t)}{1 - P_{\text{weight}}(t)} \left((24 - t_{\text{flying},h}) \cdot 10^{\log(3.1) + 0.744 \log(W(t))} + t_{\text{flying},h} \cdot 10^{\log(37.152) + 0.744 \log(W(t))} \right).$$

Then, we can solve the differential equation using $t_{\text{flying},h}$ and applying the assumptions $W(0) = 10$ and $W(365) = 35$.

7.2 Hibernating

Dragons die if their weight falls to 20% below their weight before hibernation, W_i. To model their weight after hibernation, W_f, we use a similar tactic to modeling the weight during normal activity. We model hibernation weight as

$$\frac{dW}{dt} = -\alpha \cdot E_{\text{hibernate}}.$$

Dragons hibernate for $t_{\text{hibernate}}$ years before waking according to dragon behaviors laid out in Section 5. We can integrate the weight equation over time bounds of 0 to $t_{\text{hibernate}} \cdot 365$ and weight bounds of W_i to W_f, finding

$$0.0000240024 \cdot \alpha \cdot t_{\text{hibernate}} = -0.1051423880 \cdot \frac{W_f \cdot W_i^{\frac{93}{125}} - W_i \cdot W_f^{\frac{93}{125}}}{W_i^{\frac{93}{125}} \cdot W_f^{\frac{93}{125}}}$$

For example, one of the dragons in *Game of Thrones* weighs 2,600 kg. This dragon would have to reach 2,080 kg to pass below the 20% threshold. After one year of hibernation, a dragon wakes up and hibernates again if the population of llamas still cannot support its dietary needs. This leads to potential for multiple years of hibernation in a row. Integrating the ordinary differential equation with slightly different bounds of weight between 2600 kg and 2080 kg and time between 0 and t, it would take 7.59 years of straight hibernation for that dragon to die while hibernating. A 10 kg baby would take 11.1 years of hibernation to die.

8 Caloric Intake

A dragon's daily caloric intake is based on the energy that a dragon expends during that day, E_{expended}. Dragons only eat llamas, and an average 24-month-old llama has 40 kg of meat [14], which is around 248,800 kcal per llama [15]. Then, the number of llamas consumed, L_e, is

$$L_e = \frac{E_{\text{expended}}}{248,800}.$$

The number of llamas consumed by a dragon depends on each specific dragon due to the expended energy. The developed energy equation depends on the weight of an individual dragon. If there are not enough llamas within a day's journey, then the dragon hibernates to let the llamas reproduce.

9 Llama Population

Since the dragon lives in the same cave for its adult life, we model the llama population within the $\frac{vt_{\text{flying},h}}{2}$ km-radius circle around the dragon's cave.

9.1 Initial Population

We assume llamas are evenly distributed in the Andes, which has an initial population of L_0 llamas. The length of the Andes mountain range is around 7,200 km, so creating circular regions along the length of the Andes, there are $\frac{7200}{vt_{\text{flying}h}}$ circular regions for the dragons [5]. So, a rough estimate for the initial number of llamas in each region is

$$\frac{L_0 vt_{\text{flying},h}}{7200}.$$

9.2 Change in Population

The change in llama population in one day, dL/dt, is

$$\frac{dL}{dt} = (b_r - d_r)L - \sum_{d \in \text{Dragons}} L_e,$$

with b_r being the birth rate, d_r being the death rate, and the set, Dragons being the set of dragons living in the region. A singular dragon, d, in the set carries its own individual number of llamas required to be consumed each day. The summation over all $d \in$ Dragons represents the llamas consumed by all of the dragons living in the analyzed region.

Dragon's only eat juvenile llamas according to assumption 1. The birth rate for juvenile llamas is 0.34 [4]. The death rate for juvenile llamas is 0.31, but with good handling, health, and feeding management, the death rate decreases to 0.03 [4]. Since there will be humans raising the llamas, we assume that they are taking good care of their flocks. Then,

$$\frac{dL}{dt} = 0.28L - \sum_{d \in \text{Dragons}} L_e.$$

10 Dragon Population

Let D_0 be the initial dragon population. According to *Game of Thrones* lore, these dragons can independently birth baby dragons. Thus, the population of dragons depends solely on the availability of food rather than an additional dependence on other dragons in the region. Additionally, dragons die if there are not enough llamas, but this situation is unlikely because it requires years of hibernation. This does, however, affect the overall population of dragons. The dragon population, D, is

$$D = D_0 - N_{\text{died}} + N_{\text{born}},$$

with N_{died} being the number of dragons that have died, and N_{born} being the number of dragons that have been born.

11 Applying the Model

We apply the model with $D_0 = 3$ dragons, $L_0 = 7,000,000$ llamas, $t_{\text{hibernate}} = 1$ year, $t_{\text{flying},h} = 3$ hours, $v = 200$ km/hour, and $W_r = 100$ kg, and we write a computer program to analyze what happens to dragon and llama populations overtime.

Since dragons fly three hours a day at 200 kilometers per hour, they fly 600 kilometers each day [7].

11.1 Energy Expenditure

We solve the energy expenditure differential equations using the number of hours a dragon spends flying each day.

11.1.1 Not Hibernating

Solving the differential equation

$$\frac{dW}{dt} = \alpha E_{\text{weight}},$$

describing a dragon's weight when it is not hibernating, we get

$$W(t) = \left(\frac{\left(7 \cdot 10^{\frac{2456808469}{5000000000}} + 10 \cdot 10^{\frac{284991099}{500000000}}\right) \cdot \left(-\frac{200 \ln\left(e^{-\frac{t}{200}+\frac{73}{40}}\right)}{9} - \frac{23200 \ln\left(225+109e^{-\frac{t}{200}+\frac{73}{40}}\right)}{981} + c \right) \cdot 96 \cdot \alpha}{125} \right)^{\frac{125}{32}},$$

with c being a constant.

Applying our assumption that $W(0) = 10$, we find $c = 139926.470$. We also assume that a dragon weighs 35 kg after one year, so $W(365) = 35$ and $\alpha = 0.0000548$ kg/(kcal·hr).

11.1.2 Hibernating

Solving the differential equation, we can integrate the weight equation over time bounds of 0 to 8760 hours, one year, and weight bounds of W_i to W_f, finding

$$W(t) = -0.1051423880 \cdot \frac{(W_f W_i^{\frac{93}{125}} - W_i W_f^{\frac{93}{125}})}{W_i^{\frac{93}{125}} W_f^{\frac{93}{125}}}.$$

11.2 Llama Population

Also, since each circular region has a 300-kilometer radius, half the distance a dragon flies in a day, there are 12 regions where dragons can live in the Andes. That means there are 583,333 llamas per region.

11.3 Computer Program

Our computational model simulates the dragon and llama populations per day. The computational model considers birth and death rates for both dragons and llamas as well as the dragons migrating between locations and hibernation. We consider the birth rate, death rate, hibernation, and migration of dragons to be discrete, decided at the end of each day. Hence, our model for dragon population cannot be integrated. Instead, the model is simulated via a computer program. We iterate over the days and then over the circular dragon regions.

To mathematically represent our model, we define multiple equations in Table 1.

$x(t,l)$	number of llamas with respect to time and location.
$d(t,l)$	number of dragons with respect to time and location.
$w(t,d)$	weight with respect to time and dragon.
$e(t,l)$	number of dragons that died at time and location.
$b(t,l)$	whether there exists a dragon able to give birth at a time and location.
$m(t,l)$	number of dragons that migrated at a time from a location.
$l_b(d)$	last day a dragon gave birth.
$a(d)$	age of a dragon.

Table 1: Equations used. Depending on time, location, and dragon.

Additionally, we let $L_{et} = \sum_{d \in \text{Dragons}} L_e$.

We first consider whether a dragon should currently be hibernating.

If $x(t,l) \leq L_{et}$, we initiate a cycle of hibernation that will last one year.

While hibernating, the mass of each dragon changes by

$$w(t+1,d) = w(t,d) - \alpha \cdot E_{hibernate}.$$

After each year, we calculate how many dragons have died. Consider h to be the number of years of consecutive hibernation. Then,

$$e(t,l) = \sum_{d \in \text{Dragons}} w(t,d) < 0.8w(t - 365 \cdot h, d).$$

If the condition for hibernating holds after the year, we continue for another year of hibernation.

When the dragons are not hibernating, we consider factors such as births, migrations, and regular weight gain. We begin by adjusting the weights of the dragons with

$$w(t+1,d) = \alpha E_{weight}.$$

We determine whether a dragon in the territory can give birth with

$$b(t,l) = \exists d \in \text{Dragons s.t. } (l_b(d) > 2 \cdot 365 \text{ and } w(t,d) \geq 100).$$

Next, we consider how many dragons within the territory can migrate:

$$m(t,l) = \left(\sum_{d \in \text{Dragons}} a(d) \geq 7 \cdot 365 \right) - 1.$$

We now consider the change in the dragon population at a location with

$$d(t+1,l) = \begin{cases} d(t,l) - m(t,l) - e(t,l), & \text{for } L_{et} \leq 5 \cdot x(t,l) \text{ or } \not{b}(t,l) \\ d(t,l) + 1 - m(t,l) - e(t,l), & \text{for } L_{et} \geq 5 \cdot x(t,l) \text{ and } b(t,l) \end{cases}.$$

Finally, we consider the change in the llama population with

$$x(t+1,l) = 0.28x(t,l) - L_{et} - [(d(t+1) - d(t))] \cdot \frac{503.5}{248,800}.$$

This process assumes that if we have enough llamas to justify giving birth, then we will have enough to provide the energy to give birth.

11.4 Results

The results from this model display both dragon and llama populations reaching a maximum population, the carrying capacities for both species. The llamas reach their carrying capacity and begin to oscillate between total population sizes, due to hibernation, of slightly more than 10 million to slightly more than 12 million llamas after 25 years of the dragons living on earth, as displayed in Figure 1.

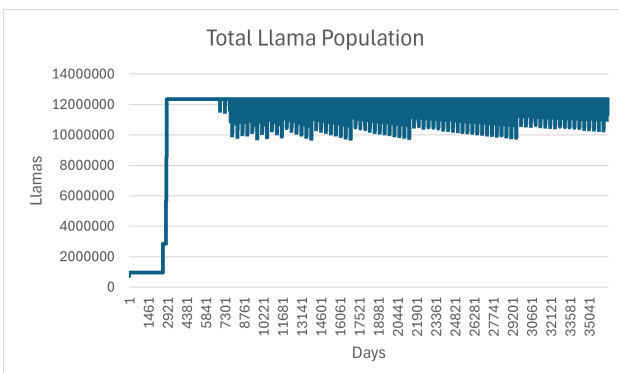

Figure 1: Llama Population over 100 Years

Figure 1 covers the llama population from the start of the simulation to the end of the 100 years. We decided to only simulate and count llama populations where dragons are present. For example, when we have one dragon, we only count the llamas within that dragon's territory, but, once we have two dragons in different territories, we count the llamas in both territories as contributing to the total. Thus, we can understand the sudden and large increase of the total llama population as multiple dragons moving to new locations within a short period of time. We also observe a pattern in the fluctuations of the llama population. For a period of time, the llama population has increasing amplitude in the dips of its population, but at some point, the amplitude of the dip resets to be a smaller value. The time it takes the population to reach the max amplitude of dips, however, becomes longer and the maximum amplitude becomes smaller. If this trend continues, we would expect periodic dips in a population of the same size to be repeated forever.

A similar plot of the dragon population from the start of the simulation to the end of the 100 years is in Figure 2. The dragon population increases smoothly until it reaches a sharp cut off in population at approximately 53,087 dragons on day 7,476, approximately year 20.5, of the simulation.

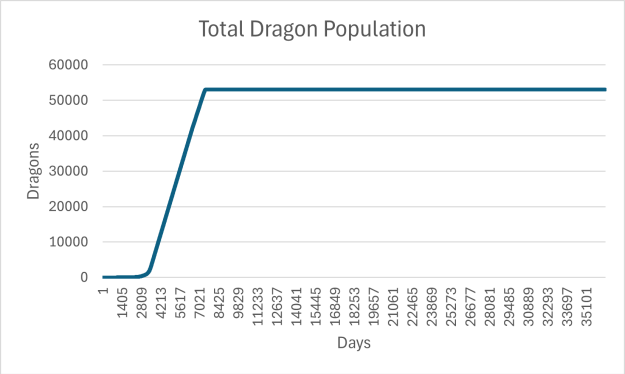

Figure 2: Dragons Population over 100 Years

This cut off corresponds to the llama population only meeting the needs of the current dragon population. Without excess food, the dragons stop reproducing and begin hibernating. A plot of the rate of reproduction with respect to time further displays the halt in reproduction after the dragons reach their carrying capacity.

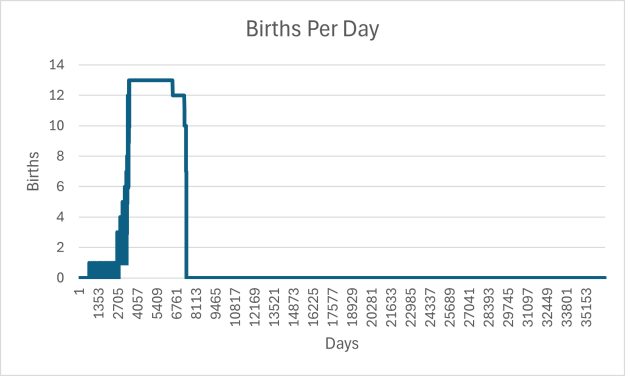

Figure 3: Number of births per day over 100 years

The number of births starts off small due to the small population of dragons existing in the area that can reproduce. As the population of dragons rises, the births per day of dragons increases to approximately 13 births per day up until the llama population can no longer support the large dragon population. With the lack of excess food, the dragons no longer birth new dragons resulting in the births quickly dropping off to zero.

The oscillations in llama populations due to the energy requirements of the increasing dragon population correspond to the hibernation of dragons in different locations. This relationship is portrayed in Figure 4.

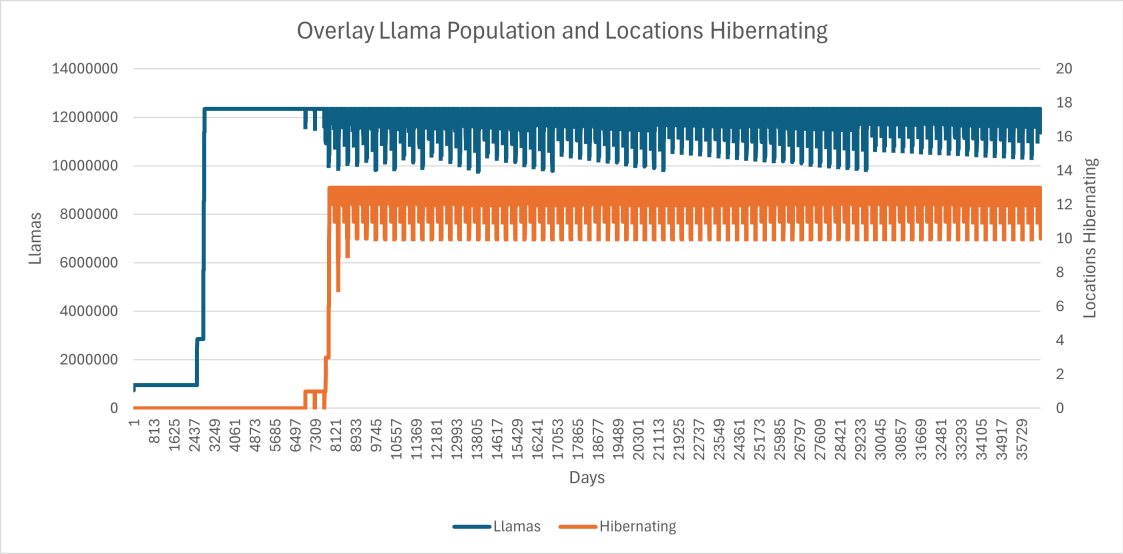

Figure 4: Llama population and Locations Hibernating over 100 years

After the llama population reaches its maximum, we observe some lag before almost all territories are in hibernation. The dragons hibernate as a group within their territory if there is insufficient food for all of the dragons in the location. The initial dips in llama population line up with the first instances of hibernation. Unlike the increasing oscillations of the llama population, the fluctuations in the total number of locations hibernating are stable past the initial increase.

Our results demonstrate that the Andes Mountain range can successfully support the existence of dragons and the large-scale expansion of their population.

11.4.1 Supporting the Dragons

Each region initially has 583,333 llamas, and dragons need support from humans to raise llamas. Each llama needs one acre, so to raise the llamas most efficiently, the humans can herd the llamas into a 583,333 acre region [16]. Using the assumption that one human can raise llamas living on 13,000 acres, the dragon would initially need 45 humans to raise the llamas.

Alternatively, llamas could remain spread out over the region, but then the dragons would need over 5,000 humans to raise the llamas because each human can only raise llamas living on 13,000 acres. Otherwise, the llama death rate could increase since the llama population would not be as healthy. Also, the dragons would need to spend more time flying to find llamas to eat, expending more energy.

If the number of llamas in the circular region reaches the carrying capacity, 1.9 million llamas, the dragons would need between 146 and 5,000 humans to raise the llamas, depending on the llama population density. Humans in the model are given a secondary purpose of putting out fires produced from dragon hunting. Since it may be hard to transport water to the fire locations, each dragon would need people to create fire lines as it breathes fire on the llamas [8]. Assuming 20 humans can create fire lines for each dragon, each dragon needs at least 65 humans to raise llamas and put out fires.

12 Further Considerations

12.1 Environmental Impacts

Large dragons living in the Andes would damage the environment more than other locations because alpine environments are fragile. Wildfires affect grasslands more than any other environment in the Andes, and the amount of area burned by wildfires has increased greatly in recent years. Since llamas live on grasslands, dragons could start fires by cooking llamas. How easily a dragon starts a wildfire depends on the climate. When it is windy and when the grasslands are drier, wildfires could start easily and be challenging to put out. The state of destruction of the ecosystem could be modeled and further applied to consider the lack of food for the llama population from the depletion of plants in the environment.

12.2 Climate Conditions

Dragons are warm-blooded, and their body temperature stays consistent via homeostasis. However, the dragons do not like the cold and they do not migrate to arctic climates, meaning these climates would not affect our model [18]. Other countries across oceans would be impossible to reach due to the limited flight distance given to the modeled dragons. These considerations limit the dragons to living in North America, Central America, or South America. Over these different locations there are different types of prey for the dragons, such as deer and sheep. The other types of prey would affect the number of humans necessary to support the dragons, and it could affect the energy a dragon expends to find prey.

We look at the climates in two cities, Ambato, Ecuador [9] and La Paz, Bolivia [10], near volcanoes in South America. The temperatures are fairly consistent throughout the year in both cities, with the average temperatures in La Paz being around 10°F cooler than those of Ambato due to the higher elevation. The decrease in temperature with increasing elevation would decrease the dragons' metabolic rates.

The average precipitation in Ambato is around 1.5 to 2.5 inches each month. La Paz has some months with a higher average precipitation. For example, La Paz experiences around 4.9 inches of precipitation in January, and the winter months in La Paz typically have less than 0.5 inches of precipitation. The varying amounts of precipitation could also affect the energy requirements and metabolic rates of dragons.

These variations in energy usage due to different climates around living areas requires analysis in a more rigorous model to more affectively comprehend the population and ecology of the scenario.

13 Other Applications

While this model is not really applicable to dragons, it provides perspective on predator-prey interactions and invasive species. Ecologists commonly use differential equation models to model the predator-prey relationships and populations of animals living in environments. The introduction of an invasive species to a region represents one significant real world application of this model. When an invasive species enters an ecosystem, it oftentimes acts as an apex predator. For example, humans accidentally introduced Burmese pythons in Florida, and without natural predators, the Burmese pythons managed to increase in population and dramatically impact the population of other animals in the Everglades [11]. Developing an apex predator model, though seemingly pointless with mythological creatures, eases the modeling of the affect that such invasive species have on local populations.

14 Conclusion

Our model indicates, it is possible for dragons to survive on Earth in the Andes Mountains and only eat llamas. We could further analyze this model by determining a practical limit on dragon weight, the maximum possible weight of the dragons based on the food resources available in their locations. There are many factors, such as limited resources and predators for young dragons, that should be considered in future models to

determine whether this smooth growth is realistic. We believe our model provides an effective estimate and tracking of hypothetical dragon and llama populations, but for the more practical use, it should expand to allow inputs from the user so that they can model their dragons or other animals as they wish.

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